

Higher dimensions

We continue applying the principle of looking at relationships between things, because we also need to look at relationships between those relationships, and so on. This gives us more dimensions, possibly infinitely many.

In this final chapter I'll give a broad sweeping overview of my particular field of research: higher-dimensional category theory. This will not be detailed or rigorous; the aim is to give an idea and flavor of how the field develops into higher dimensions. Rigorous details are far beyond our scope, but I hope the ideas feel like a natural development of all we've done so far.

24.1 Why higher dimensions?

We began our journey into category theory with the idea of relationships. We said that we wanted to study things in the context of their relationships with other things, rather than in isolation. This gave us the idea of morphisms between objects; taking objects and morphisms together is what gave us the notion of a category. This gave us, among other things, a more nuanced notion of sameness, and a way of studying various kinds of abstract structure arising from patterns in relationships.

But now, if we believe in studying relationships between things, what about relationships between morphisms? What about a more nuanced notion of sameness between morphisms? What about a way of studying various kinds of abstract structure on morphisms? If we're interested in categories rather than sets, how about categories of morphisms instead of sets of them?

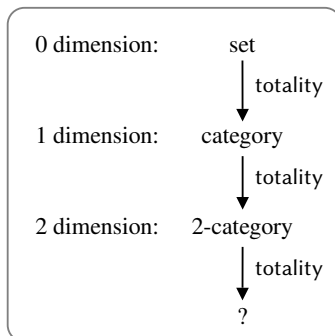
This is a way in which abstract principles nudge us into higher dimensions just by us using our imagination. A way in which something more like practice nudges us that way is if we think about totalities of structures. We have seen that sets are 0-dimensional structures, as they just consist of points; but sets and functions together make a category, which is 1-dimensional.

Then, categories come with functors and also natural transformations. The natural transformations are "morphisms between functors" so they are one di-

mension higher, and they arise from the existence of morphisms *in* categories. So the extra dimension *inside* categories produced an extra dimension in the *totality* of categories.

When we have objects, morphisms, and morphisms between morphisms, this is called a 2-category. For more succinct terminology we sometimes call these 0-cells, 1-cells and 2-cells. This terminology also generalizes well into more dimensions (rather than saying morphisms between morphisms between morphisms between. . .).

We have a growing system of structures and their totalities, as shown on the right. This definitely suggests that things are not going to stop at 2-categories. As you might have guessed, the totality of 2-categories is a 3-category, and this continues so that we have a notion of n -category for every finite n , and n -categories form an $n + 1$ -category.



More specifically, if we look at a totality of 2-categories it seems quite likely that we will gain an extra level of structure beyond functors and natural transformations. Natural transformation arose from the existence of morphisms in categories, but now we also have 2-cells. The 2-cells *inside* a 2-category go on to produce morphisms between natural transformations.

We are also nudged towards higher-dimensional thinking just by dreaming of morphisms between morphisms between morphisms, and so on, for dimension after dimension. In fact, why do we ever need to stop? The answer is we don't. In an n -category we have morphisms at every dimension up to n , but we could have a higher-dimensional structure with morphisms at every dimension "forever", with no upper limit. This is called an *infinity category*.[†]

This is all very well as an idea, but just like most dreams it is much harder to realize in practice than just to dream about. In this chapter we're going to sketch out how to do it, why it's hard, and why therefore there are reasons to try and keep things finite-dimensional if we can.

This is the last chapter in the book and is about a huge open research field, so much of this will be a sketch and a glimpse rather than anything rigorous. I will include more Things to Think About that I will not fully write out afterwards.

[†] They are sometimes also called ω -categories, where ω is the Greek letter omega which is often used to denote the "first" infinity in set theory, that is, the size of the natural numbers. We use this because really we're saying there is a dimension of morphism for every natural number.

24.2 Defining 2-categories directly

As a starting point we can look at the full definition of 2-category, which we have hinted at. Many of the issues with higher dimensions can be illustrated in 2 dimensions, including the fact that there are many different possible approaches and they diverge more and more as dimensions increase.

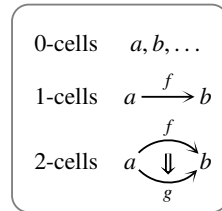
For the definition of a 2-category we basically take the structure of categories, functors and natural transformations and encapsulate it as data, structure and properties. The most elegant ways to do it require some more theory, but here is a direct definition first.

2-categories: elementary definition

A 2-category $\mathcal{A}$ consists of:

DATA

- A collection of 0-cells.
- For any 0-cells a, b a collection $\mathcal{A}(a, b)$ of 1-cells from a to b , drawn as shown.
- For any parallel 1-cells $f, g: a \rightarrow b$ a collection $\mathcal{A}(f, g)$ of 2-cells from f to g , drawn as shown.



STRUCTURE

- Identity 1-cells: $\forall$ 0-cells a a 1-cell $a \xrightarrow{1_a} a$
- Composition of 1-cells: $\forall$ 1-cells $a \xrightarrow{f} b \xrightarrow{g} c$ a 1-cell $a \xrightarrow{g \circ f} c$
- Identity 2-cells: $\forall$ 1-cells $a \xrightarrow{f} b$ a 2-cell $a \Downarrow_{1_f} b$
- Composition of 2-cells
 - vertical:[†] $\forall$ 2-cells $a \xrightarrow{f} b \xrightarrow{g} b$ a 2-cell $a \Downarrow_{\beta \circ \alpha} b$
 - horizontal: $\forall$ 2-cells $a \xrightarrow{f} b \xrightarrow{h} c$ a 2-cell $a \Downarrow_{k \circ g} c$

PROPERTIES

- Unit laws: the identity 1-cells act as identities with respect to 1-cell composition, and identity 2-cells act as identities with respect to vertical composition of 2-cells.

[†] In vertical composition, the 1-cell g isn't there in the composite because it's like the middle object — the composite goes all the way from end to end.

- Associativity: composition of 1-cells, and both horizontal and vertical composition of 2-cells are associative.
- The interchange law (Section 22.10), summed up in this diagram:

$$\begin{array}{ccc} \begin{array}{c} \downarrow \quad \downarrow \\ \hline \downarrow \quad \downarrow \end{array} & = & \begin{array}{c} \downarrow \quad \downarrow \\ \hline \downarrow \quad \downarrow \end{array} \end{array}$$

Perhaps you feel that this definition has a “nuts and bolts” feel to it, and that it doesn’t really have a satisfying logic behind it; if so then you are thinking like a category theorist. I will now sketch two more abstractly elegant ways of approaching this definition, both of which come to the same thing when unraveled. When we are making abstract definitions there is no real concept of right and wrong, as long as we don’t actually cause a contradiction. There are instead guiding principles:

- External: are there many examples of this structure?
- Internal: is there some compelling internal logic behind the definition?

The nuts and bolts definition came from looking at a motivating example: the totality of categories, functors and natural transformations. We’ll now try two ways of looking at the internal logic, coming from the two approaches to defining categories in the first place: by homset, or by underlying graph.

24.3 Revisiting homsets

In this approach we are going to take seriously the idea of thinking about relationships between morphisms. Instead of having *sets* of morphisms, we are going to have *categories* of morphisms. Then any part of the definition that was previously a function between sets of morphisms is now going to have to be a *functor* between categories of morphisms. This is a general abstract process called *enrichment*: we “enrich” a category by allowing its morphisms to have some structure on them. As we’ll see, it is a formal definition, not just an idea.

First we need to revisit the definition of category in an abstract way that best suits this generalization. The following expression is the same definition we’re used to, just expressed slightly more “categorically”, that is, using sets and functions rather than sets and elements.

Definition of category: revisited

A (locally small) category $\mathcal{C}$ is given by

DATA

- A collection of objects.
- For any objects a, b a set $\mathcal{C}(a, b)$ of morphisms from a to b .

STRUCTURE

- Identities: for any object a , a function as shown,[†] picking out an identity morphism $a \xrightarrow{1_a} a$.

$$\begin{array}{ccc} 1 & \xrightarrow{\text{id}} & \mathcal{C}(a, a) \\ * & \mapsto & 1_a \end{array}$$

- Composition: for any objects a, b, c , a function as shown, sending composable pairs to composites.

$$\begin{array}{ccc} \mathcal{C}(b, c) \times \mathcal{C}(a, b) & \xrightarrow{\text{comp}} & \mathcal{C}(a, c) \\ b \xrightarrow{g} c, a \xrightarrow{f} b & \mapsto & a \xrightarrow{g \circ f} c \end{array}$$

For the last part we need to recall that the product of two sets contains ordered pairs with one element from each set. So the product $\mathcal{C}(b, c) \times \mathcal{C}(a, b)$ is a set containing pairs of morphisms, one with target b and one with source b , so they are composable at b . The reason we've put it in the “backwards” order is to match up with the “backwards” order in which we write composition.

PROPERTIES

The axioms for units and associativity can now be expressed entirely according to these *functions* on homsets, without reference to *elements* of homsets.

The two triangles are the unit axioms saying that identity morphisms act as identities on both the left and the right; the square underneath is associativity.

I've put the diagrams here for you to see, but won't go into much explanation; you might like to chase elements around them and see if you can understand why they correspond to the axioms in question.

$$\begin{array}{ccc} \mathcal{C}(a, b) \cong 1 \times \mathcal{C}(a, b) & \xrightarrow{\text{id} \times 1} & \mathcal{C}(a, a) \times \mathcal{C}(a, b) \\ & \searrow 1 & \downarrow \text{comp} \\ & & \mathcal{C}(a, b) \\ \\ \mathcal{C}(a, b) \cong \mathcal{C}(a, b) \times 1 & \xrightarrow{1 \times \text{id}} & \mathcal{C}(a, b) \times \mathcal{C}(b, b) \\ & \searrow 1 & \downarrow \text{comp} \\ & & \mathcal{C}(a, b) \\ \\ \mathcal{C}(c, d) \times \mathcal{C}(b, c) \times \mathcal{C}(a, b) & \xrightarrow{1 \times \text{comp}} & \mathcal{C}(c, d) \times \mathcal{C}(a, c) \\ \downarrow \text{comp} \times 1 & & \downarrow \text{comp} \\ \mathcal{C}(b, d) \times \mathcal{C}(a, b) & \xrightarrow{\text{comp}} & \mathcal{C}(a, d) \end{array}$$

Now that we are only referring to homsets and functions between them (not elements), we have made the definition more “categorical”. All the diagrams above are just diagrams in the category **Set**, so we can take this definition and try to put it inside any other category $\mathcal{V}$. This means that everywhere we

[†] Here 1 denotes the terminal set; remember a function from the terminal set just picks out an element of the target set. I've written $*$ for the single object of the terminal set.

referred to a set of morphisms we now have an object of $\mathcal{V}$ instead, and everywhere we have a function between homsets we now have a morphism in $\mathcal{V}$. If you look at the definition carefully you will see that at one point we refer to products of homsets, so actually it looks like we need products in $\mathcal{V}$. It turns out that something slightly less stringent than a categorical product will do, called a monoidal structure. But we'll come back to that.

If we move all our homsets from **Set** to some category $\mathcal{V}$, this is the notion of a *category enriched in $\mathcal{V}$* . We can then define 2-categories as categories enriched in **Cat**. This means that instead of homsets we have hom-categories, and structure is now given by functors between hom-categories. We will now sketch out that definition.

Definition of 2-category by enrichment

A (locally small) 2-category $\mathcal{C}$ is a category enriched in **Cat**. That is:

DATA

- A collection of objects.
- For any objects a, b a category $\mathcal{C}(a, b)$ of morphisms from a to b .

Note that the objects and morphisms of this hom-category are then the 1-cells and 2-cells of the 2-category. Composition in this category is vertical composition.

STRUCTURE

- Identities: for any object a a functor $\mathbb{1} \rightarrow \mathcal{C}(a, a)$ picking out an identity 1-cell $a \xrightarrow{1_a} a$.

Note that $\mathbb{1}$ is the terminal category, with one object and just the identity morphism. The identity morphism in $\mathbb{1}$ has to be sent to the identity morphism in $\mathcal{C}(a, a)$ so we get no further information from looking at the action on morphisms.

- Composition: for any objects a, b, c a functor

$$\begin{array}{ccc}
 \mathcal{C}(b, c) \times \mathcal{C}(a, b) & \longrightarrow & \mathcal{C}(a, c) \\
 \text{on objects (1-cells):} & b \xrightarrow{g} c, \quad a \xrightarrow{f} b & \longmapsto a \xrightarrow{g \circ f} c \\
 \text{on morphisms (2-cells):} & \begin{array}{c} b \begin{array}{c} \xrightarrow{g_1} \\ \Downarrow \beta \\ \xrightarrow{g_2} \end{array} c, \quad a \begin{array}{c} \xrightarrow{f_1} \\ \Downarrow \alpha \\ \xrightarrow{f_2} \end{array} b \end{array} & \longmapsto a \begin{array}{c} \xrightarrow{g_1 \circ f_1} \\ \Downarrow \beta * \alpha \\ \xrightarrow{g_2 \circ f_2} \end{array} c
 \end{array}$$

So the composition functor gives us both composition of 1-cells and horizontal composition of 2-cells.

PROPERTIES

The axioms for units and associativity are now the same commutative diagrams we drew for the categorical definition of a category, they just happen to live in **Cat** now rather than **Set**.

Things To Think About

T 24.1 For me the most interesting thing about this definition is the fact that functoriality of the composition functor corresponds to the interchange law. See if you can unravel that.

Note that once we start going into higher dimensions we may feel a compulsion to keep asking about the next dimension. With the definition of category enriched in $\mathcal{V}$ we can iterate the process as long as we can always form products. Then 2-categories will form a category with products, and if we enrich in that we'll get 3-categories, and so on.

You might notice that 2-categories are really supposed to form a 3-category. That's a good thing to notice, and is one of the reasons things get more subtle. You might also have noticed that we blithely said we would ask for some commutative diagrams of functors. This is not really the right level of sameness as it involves equality of functors, and we now know that we could (and perhaps should) ask for them to be naturally isomorphic instead. This is the question of strictness and weakness, which we'll come back to shortly.

24.4 From underlying graphs to underlying 2-graphs

A different way to go up a dimension starts from the definition of a (small) category via an underlying graph, as shown here.

$$C_1 \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} C_0$$

As this is a diagram in **Set**, we could try putting the entire diagram inside **Cat** instead of **Set**. This is called *internalization* but it gives us something slightly different: we'll get a category of morphisms and a category of objects as well. We will come back to that.

For now we'll directly extend the graph by one dimension, to include a set of 2-cells like this:

$$C_2 \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} C_1 \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} C_0$$

However we now need a condition on the source and target maps, to ensure that our 2-cells are globular shaped, as shown here.

$$\begin{array}{ccc} & f & \\ a & \Downarrow \alpha & b \\ & g & \end{array}$$

This shape occurs because the source and target 1-cells of α themselves have the same source and target. That is $ss = st$ and $ts = tt$. This is called the *globularity condition*. A diagram of sets and functions satisfying this condition

is called a (2-dimensional) *globular set* or 2-globular set[†] and generalizes into n dimensions. Our original graphs are 1-globular sets.

Things To Think About

T 24.2 A 2-globular set has three key sub-1-globular sets; can you find them?

Here are the three most interesting 1-globular sets that live inside the above 2-globular set. The first two are somewhat obvious, involving just the right-hand part or the left-hand part; the third comes from composing all the way from end to end.

$$\begin{aligned} C_1 &\xrightarrow[s]{t} C_0 \\ C_2 &\xrightarrow[t]{s} C_1 \\ C_2 &\xrightarrow[ts=tt]{ss=st} C_0 \end{aligned}$$

Things To Think About

T 24.3 In the case of the 2-category[‡] of categories, functors and natural transformations, what does each of these 1-globular sets represent? Can you verify that each one is a category?

In the 2-category of categories, functors and natural transformations, the first sub-globular set (collection) above has categories as its objects, and functors as its morphisms. We know this has the structure of a category: it's the basic category **Cat** from before we even thought about natural transformations.

The second sub-globular set (involving C_2 and C_1) has functors as objects and natural transformations as morphisms. This also forms a category. Previously we have defined functor categories $[\mathcal{C}, \mathcal{D}]$ only after fixing the source and target categories, but we can throw all functors together into a large category of functors and natural transformations with any endpoints.

The third sub-globular set is a little curious. It considers natural transformations with respect to their source and target *categories* rather than the functors.

So the natural transformation shown on the right is considered to have source $\mathcal{A}$ and target $\mathcal{B}$.

$$\begin{array}{ccc} & F & \\ \mathcal{A} & \Downarrow \alpha & \mathcal{B} \\ & G & \end{array}$$

We will then consider composition when the target category of one natural transformation matches the source category of another, that is, horizontal composition.

$$\begin{array}{ccccc} & F & & H & \\ \mathcal{A} & \Downarrow \alpha & \mathcal{B} & \Downarrow \beta & \mathcal{C} \\ & G & & K & \end{array}$$

So asserting that each of these three sub-globular sets is a category gives us, respectively, composition of 1-cells, vertical composition of 2-cells, and horizontal composition of 2-cells. In this framework we have to assert interchange separately, but we essentially have the following definition.

[†] We could also call it a 2-graph but technically that is defined slightly differently.

[‡] That is not actually a *small* 2-category but the ideas still hold, just with collections instead of sets.

2-categories: globular set definition

A (small) 2-category is a 2-globular set in which

- every sub-1-globular set has the structure of a category, and
 - interchange holds.
-

This definition gives the same structure as the one defined by enrichment (aside from size issues) but it has a different feel to it, and leads to different types of generalization. We saw that even for ordinary categories these two different characterizations (by homset, or by underlying graph) give us slightly different insights. For example, from the graph approach we saw how duality naturally arises from the symmetry between s and t . In a 2-globular set we have *two* pairs of source/target functions, each with analogous symmetry. We can flip either pair independently. Flipping the left-hand pair consists of flipping the direction of the 2-cells; flipping the right-hand pair consists of flipping the direction of the 1-cells; or we can flip both.

- Flipping the direction of the 1-cells is like what we do with ordinary categories, and this gets called “op” just like for categories.
- Flipping the direction of the 2-cells means we’re taking the duals of the hom-categories; this gets called “co”.
- Flipping both means we flip everything, and is inevitably called “co-op”.

It recently occurred to me to try and think of the opposite of “FOMO” (the Fear Of Missing Out). During the COVID-19 era I was very grateful not to have to join in with anything, being lucky to be able to work from home. I thought of calling it “FOJI” for Fear Of Joining In, but then I saw that others had called it “JOMO” for Joy Of Missing Out. Those are not quite the same thing, just as the op and the co of a 2-category are not in general the same thing. I decided that if this were a higher-dimensional structure then the emotion (fear/joy) would be a higher dimension than the thing itself (missing out/joining in), so

- $(\text{FOMO})^{\text{op}} = \text{FOJI}$
- $(\text{FOMO})^{\text{co}} = \text{JOMO}$
- $(\text{FOMO})^{\text{coop}} = \text{JOJI}$ — Joy Of Joining In.

I think JOJI is really not the same as FOMO; the latter seems a little sad, whereas JOJI is a much more active and positive idea.

That is all very far from rigorous but is the sort of way one’s brain might start working when one spends a lot of time thinking about higher dimensions. (Well, mine does.)

Once we know what 2-categories are, there are various natural things to do

to develop category theory. We could develop “2-category theory”: take everything we’ve done in one dimension and try to do it in two, by replacing equalities with isomorphisms and working out what axioms those isomorphisms should satisfy one dimension up.

Another key way to progress is to express everything 2-categorically. Recall that one of the principles of 1-category theory was to take a “familiar” structure involving sets, and express it just using sets and *functions* rather than sets and elements. We then go round looking for the structure in categories other than **Set**. Once we know about 2-categories we can take a familiar structure involving categories, and express it just using categories, functors and natural transformations (rather than objects and morphisms inside any categories). We can then go round looking for it in 2-categories other than **Cat**.

Things To Think About

T 24.4 Think about the two definitions of equivalence of categories. Which one is 2-categorical? Can you see how this gives us a notion of sameness for 0-cells in any 2-category, slightly weaker than isomorphism?

The first definition we saw was “pointwise” equivalence of categories. The definition was expressed in terms of a functor being full, faithful and essentially surjective on objects. The phrase “on objects” is a clue to this involving elements, as is the term “pointwise”; note that the definitions of full and faithful involve thinking about morphisms *inside* categories, rather than just functors *between* them.

The second definition of equivalence of categories was expressed in terms of pseudo-inverses: functors F and G going back and forth, and then natural isomorphisms $GF \cong 1$ and $FG \cong 1$. As this definition only involves categories, functors and natural transformations between them, it is something we can now do in any 2-category instead of **Cat**. We would just express it using 0-cells, 1-cells and 2-cells, instead of categories, functors and natural transformations.

This gives us the “correct” notion of sameness for 0-cells in a 2-category. Isomorphism is too strong as it invokes equalities between 1-cells. However, for 1-cells isomorphism is the “correct” notion of sameness as it only invokes equalities between 2-cells; 2-cells are the top dimension, so equality is the only relationship we have for them.

In summary, here are the “correct” notions of sameness for cells at each dimension in a 2-category.

dimension	sameness
0-cells	equivalence
1-cells	isomorphism
2-cells	equality

We see that the more dimensions that a cell has *above* it, the more subtle a notion of sameness we can get. The number of dimensions that a cell has above it is called its *codimension*.

So far the only example of a 2-category we have seen is the 2-category of categories, functors and natural transformations. We will now look at an important special case of 2-categories: those with only one 0-cell. These are called monoidal categories.

24.5 Monoidal categories

Monoidal categories are one of the most widespread examples of a *slightly* higher-dimensional structure. They are a “categorification” of the concept of monoids. Categorification[†] is the process of going up a dimension by turning sets into categories. It is not exactly a rigorously defined term, because it is not a straightforwardly definable process. However, it is a good guiding principle.

A monoid is a set equipped with a binary operation; the categorification will be a category equipped with a binary operation, and is called a monoidal category. So far this is an idea, not a definition. The abstract framework of 2-categories helps us to make the definition in a way that we know will have good internal logic. We can proceed by analogy with monoids. Monoids arise as one-object categories, and when we unravel this we find that the non-trivial information amounts to a set equipped with a binary product. We can proceed analogously, but this time starting from 2-categories, as follows.[‡]

Definition 24.1 A *monoidal category* is a 2-category with only one 0-cell.

If we unravel this we can perform a dimension shift just like we did for monoids:

- The single 0-cell gives us no information so we can ignore it.
- We have non-trivial 1-cells and 2-cells; we regard these as objects and morphisms in the new structure.
- All 1-cells are now composable, so $\circ$ on 1-cells becomes a binary operation on our new objects, which we often write as $\otimes$.
- All 2-cells are now horizontally composable so $*$ on 2-cells becomes a binary operation, that is, $\otimes$ extends to morphisms in our new structure.

[†] This term was coined by Louis Crane.

[‡] The original definition was made directly as a category with a binary operation, just as the original definition of monoid was made directly as a set with a binary operation.

Here is a diagram of the dimension shift:

2-category	----->	monoidal category
unique 0-cell	----->	
1-cells	----->	objects
2-cells	----->	morphisms
composition of 1-cells	----->	$\otimes$ on objects
horizontal composition of 2-cells	----->	$\otimes$ on morphisms
vertical composition of 2-cells	----->	composition of morphisms

We can think about this more abstractly using the enriched definition of a 2-category, in which we have hom-categories $\mathcal{C}(a, b)$ for any objects a, b .

For any objects a, b, c we have a composition functor as shown here, giving composition of 1-cells and horizontal composition of 2-cells.

$$\mathcal{C}(b, c) \times \mathcal{C}(a, b) \longrightarrow \mathcal{C}(a, c)$$

Now if our 2-category has only one 0-cell, say $*$, then there is just one hom-category $\mathcal{C}(*, *)$ and a single composition functor:

$$\mathcal{C}(*, *) \times \mathcal{C}(*, *) \longrightarrow \mathcal{C}(*, *)$$

The “dimension shift” then consists of not bothering to think about the single object $*$, and writing the single hom-category $\mathcal{C}(*, *)$ simply as $\mathcal{C}$. This is the underlying category of our monoidal category.

The composition functor then becomes a functor as shown here. We often write it as $\otimes$, pronounced “tensor”, and it’s not to be confused with composition $\circ$.

	$\mathcal{C} \times \mathcal{C}$	$\xrightarrow{\otimes}$	$\mathcal{C}$
on objects:	a, b	$\longmapsto$	$a \otimes b$
on morphisms:	$\begin{array}{c} a & b \\ f \downarrow & \downarrow g \\ a' & b' \end{array}$	$\longmapsto$	$\begin{array}{c} a \otimes b \\ \downarrow f \otimes g \\ a' \otimes b' \end{array}$

Things To Think About

T 24.5 Earlier on I said I found it interesting that functoriality of the composition functor in a 2-category gives interchange; if you like, try and see what the functoriality of this $\otimes$ functor does.

There are further categorified analogies we can make with monoids. Recall that we can think of the binary operation in a monoid as being like a form of multiplication, but it need not actually come from multiplication.

Likewise for monoidal categories we can think of the $\otimes$ operation as being

like a multiplication, and while it can come from categorical products (with some caveats about weakness which we'll come to) it need not. We mentioned this when we gave the definition of enriched categories: we can enrich in a category with products, but we can also do it in a monoidal category where the $\otimes$ operation isn't a categorical product.

One motivating example for abstract mathematicians is the tensor product of vector spaces; tensor products in general are subtle binary operations on abstract structures, which are not the same as products but can be studied a little like products. The framework of monoidal categories gives us a way to do that. As tensor products arise in a vast range of mathematical fields, monoidal categories are one of the most widespread examples of categorical structures.

The main subtlety comes from the issue of what the axioms should be. This is the question of weakness.

24.6 Strictness vs weakness

We have not been terribly rigorous in our exploration of 2-categories, but you might have noticed that we did not do things as subtly as we might: we asked for some equalities at levels where something more nuanced was possible. This is the issue of “weakness” for higher-dimensional categories. If we ask for strict equalities even though a more nuanced type of sameness is available, we are doing things *strictly*. So far we have really been talking about *strict* 2-categories. To do things *weakly*, we must use the most subtle version of sameness available at whatever dimension of cell we're looking at.

In the definition of 2-category the issue arises where we asked for diagrams to commute in **Cat**. Those diagrams gave us the unit and associativity laws, but were at the “wrong” level of weakness: the more nuanced notion of sameness for functors is natural isomorphism. So we did something unnecessarily strict.

It is a little easier to see with monoidal categories.

Things To Think About

T 24.6 Suppose we have a category $\mathcal{C}$ and a functor $\otimes$ giving us a binary operation on objects and on morphisms. What would associativity look like? What's wrong with that?

Associativity on objects would look like this: $(a \otimes b) \otimes c = a \otimes (b \otimes c)$.

This is an equality between objects in a category, so it's the “wrong” level of sameness. The correct level is an isomorphism. And the best thing to do is not just to say that these things are isomorphic, but to specify an isomorphism showing it, like this: $(a \otimes b) \otimes c \xrightarrow{\alpha_{abc}} a \otimes (b \otimes c)$.

Things To Think About

T 24.7 What would the isomorphisms look like for the unit laws?

For the unit laws the same thought process leads us to realize that we don't want the identity to act as a strict identity but as a weak one. As a result we tend not to write it as 1 any more, but perhaps I . The unit isomorphisms look like this: $a \otimes I \xrightarrow{r_a} a$ and $I \otimes a \xrightarrow{l_a} a$. The letters r and l stand for "right" and "left" as they're dealing with a unit on the right or a unit on the left.

The "special" isomorphisms that we specify to replace the equalities for associativity and unit laws are generally called "structure isomorphisms" or "coherence isomorphisms". There are now several more complications:

- What happens at the level of morphisms?
- How are the structure isomorphisms for different objects related?
- Do the structure isomorphisms satisfy some axioms?

The general idea now is that "everything that could commute does commute". That is, whenever there are two ways to get from one place to another using the coherence isomorphisms, they should give the same answer so that we don't get abstract anarchy.

Things To Think About

T 24.8 Try working out what happens for associativity of $\otimes$ on morphisms. You might think we still have to ask for strict associativity on morphisms, because that is the top level of dimension. However, if you draw out the endpoints you should find that that would not typecheck. What structure isomorphisms could you insert to fix that? If you can do that, see if you can "follow your abstract nose" to write down some other things you might like to be true.

Here is a case where writing down strings of symbols could lead us astray. We could merrily write down an axiom for associativity of morphisms, like what we have in the strict case: $(f \otimes g) \otimes h = f \otimes (g \otimes h)$. However this doesn't typecheck any more, as we'll now see.

Suppose the original morphisms are f , g , and h as shown here; I've drawn them vertically as it might help when we start doing $\otimes$ on them.

$$\begin{array}{ccc} a & b & c \\ \downarrow f & \downarrow g & \downarrow h \\ a' & b' & c' \end{array}$$

Then the two sides of the supposed associativity equation on morphisms are the morphisms shown here.

$$\begin{array}{ccc} (a \otimes b) \otimes c & & a \otimes (b \otimes c) \\ \downarrow (f \otimes g) \otimes h & & \downarrow f \otimes (g \otimes h) \\ (a' \otimes b') \otimes c' & & a' \otimes (b' \otimes c') \end{array}$$

The issue is that in a weakly associative situation those endpoints don't match, so we can't ask for the morphisms to be equal.

However, we do have structure isomorphisms mediating between the ends, so we can ask instead for this diagram to commute.

$$\begin{array}{ccc}
 (a \otimes b) \otimes c & \xrightarrow{\alpha_{abc}} & a \otimes (b \otimes c) \\
 \downarrow (f \otimes g) \otimes h & & f \otimes (g \otimes h) \downarrow \\
 (a' \otimes b') \otimes c' & \xrightarrow{\alpha_{a'b'c'}} & a' \otimes (b' \otimes c')
 \end{array}$$

You might notice that this looks a bit like a naturality square. In fact, it looks so much like a naturality square, it *is* a naturality square. The structure isomorphisms α_{abc} compile into a natural transformation, as do the ones for units.

We still need some axioms for the structure isomorphisms. We want them to behave well enough that we can manipulate things somewhat as if they were strict equalities, even though they're not. That means that if we move parentheses around using the isomorphisms, it shouldn't matter in what steps we do it — one of the key things about equalities is that we can pile them up on each other, and keep substituting things for other things safe in the knowledge that the equalities will still hold. This is what we want here.

We can look for possible issues by thinking about situations in which different paths might have been possible, and then ask for them to be equal. For example, if we're faced with the expression $(a \otimes b) \otimes c$, there is only one way to use a structure isomorphism to move the parentheses to the right. However with the expression $((a \otimes b) \otimes c) \otimes d$, there are two things we could do.

Things To Think About

T 24.9 Can you find the two ways to move parentheses to the right in the expression $((a \otimes b) \otimes c) \otimes d$, using structure isomorphisms? Once you've done the first step (in two ways), you can keep moving parentheses to the right without any further branching choices. This makes two paths built from structure isomorphisms. Follow them around until they meet up again; those will be two paths that we want to ensure compose to the same thing.

We get this pentagon called the *associativity pentagon*; it is one of the axioms for a weak monoidal category. The other axiom involves the unit object I .

$$\begin{array}{ccc}
 & (a \otimes (b \otimes c)) \otimes d & \longrightarrow a \otimes ((b \otimes c) \otimes d) \\
 \nearrow & & \searrow \\
 ((a \otimes b) \otimes c) \otimes d & & a \otimes (b \otimes (c \otimes d)) \\
 \searrow & & \nearrow \\
 & (a \otimes b) \otimes (c \otimes d) &
 \end{array}$$

Things To Think About

T 24.10 Can you find two ways to use structure isomorphisms to “get rid of the I ” in the expression $(a \otimes I) \otimes b$? Again, when the arrows meet up again we get a diagram that we will ask to commute.

We get this triangle, which is called the *unit triangle*, and is the other axiom for a weak monoidal category.

$$\begin{array}{ccc}
 (a \otimes I) \otimes b & \xrightarrow{\alpha_{alb}} & a \otimes (I \otimes b) \\
 \searrow r_a \otimes 1_b & & \swarrow 1_a \otimes I_b \\
 & a \otimes b &
 \end{array}$$

The pentagon and the triangle are the axioms we need for a weak monoidal category, to ensure that we can manipulate structure isomorphisms as if they were equalities. In fact, as this is a one-object 2-category, those are the axioms we need for a weak 2-category, also called a *bicategory*. The structure isomorphisms are also called *coherence constraints*, and the question of how constraints in general interact is one of the big questions of higher-dimensional category theory. It is the question of coherence.

24.7 Coherence

Coherence questions in category theory are essentially about which diagrams commute in a particular type of category, as a consequence of the axioms. At least, that's what they're about on the surface. Deep down they're about the interaction between different ways of presenting the structure in question. We briefly mentioned in Chapter 8 the two different ways of presenting basic associativity (of multiplication or composition, say):

1. Local: for any a, b, c , $(ab)c = a(bc)$.
2. Global: any finite string $a_1 a_2 \cdots a_n$ has a unique well-defined product.

For monoidal categories we have something analogous:

1. Local: for any objects a, b, c, d the above pentagon and triangle commute.
2. Global: all diagrams of structure isomorphisms commute.

The point of the “global” description is that if all diagrams of structure isomorphisms commute, then we can move around them without worrying exactly which path we took, which is somewhat how we manipulate equalities.

The fact that those two presentations of monoidal categories (or bicategories) are equivalent is a deep theorem of Mac Lane called a *coherence theorem*, specifically, *coherence for monoidal categories*. Coherence typically involves a question of which presentation is the definition and which one is given as a theorem. This is a question for ordinary associativity as well as for higher-dimensional structures. Monoidal categories are typically defined according to the two axioms, and then we prove a theorem saying that “all diagrams commute” (this has to be made precise of course, so this is more of a slogan than a theorem). The point is to find a small set of axioms that generates all the commuting diagrams. However, another point of view is to define monoidal

categories by saying “all diagrams commute” and then prove a coherence theorem saying that it suffices to know that the pentagon and triangle commute.

The two presentations have different uses in practice. When we are checking that something is a monoidal category in the first place it is necessary to have a small set of axioms to check because we can’t physically check that all diagrams commute. But when we are working with (or in) a monoidal category we usually use the fact that all diagrams commute, as the diagram we’re looking at is typically not precisely a pentagon or triangle. The important thing is that we have both possibilities and we know they’re equivalent.

The small set of generating axioms can seem a little arbitrary sometimes, and I personally think this is one of the reasons abstract math can seem baffling. By contrast, the global presentation makes sense in some fundamental way. As a case in point, when Mac Lane was first making the definition of monoidal category his list of axioms was longer; it was only later that Max Kelly proved that some were redundant. However, arguably Mac Lane had a point, because more axioms turn out to be needed to understand the next dimension up properly.

Another way of interpreting coherence for monoidal categories is that weakness doesn’t matter *that* much, and we can more or less behave as if it’s a strict monoidal category without anything going too wrong. This is why, for example, we can take cartesian products of three sets and not worry too much whether we’re really taking ordered triples (a, b, c) or $((a, b), c)$ or $(a, (b, c))$.

There are many ways to state that coherence theorem and they’re all beyond our scope, but here’s one for the record anyway.

Theorem 24.2 (Coherence for monoidal categories) *Every weak monoidal category is monoidal equivalent to a strict one.*

As I hope you can guess by now, “monoidal equivalent” is the correct notion of sameness for a monoidal category, and is essentially an equivalence of categories where the functors in question respect the monoidal structure. This coherence theorem is a one-object version of the one for weak 2-categories, which are also called bicategories:

Theorem 24.3 (Coherence for bicategories) *Every bicategory is biequivalent to a (strict) 2-category.*

Sometimes it’s easier to think about monoidal categories just because we don’t have to draw so many dimensions. Studying higher-dimensional structures in which the bottom dimensions are trivial (containing only one cell) can help us understand and deal with higher dimensions at many levels. A structure with trivial lower dimensions is called *degenerate*.

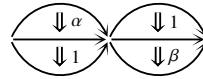
24.8 Degeneracy

A degenerate category is one where the lowest dimension is trivial, that is, there is only one object. Once we're in a 2-category there is one further level of degeneracy we could consider. We have seen that a 2-category with just one 0-cell "is" a monoidal category, but what about a 2-category with just one 0-cell and also one 1-cell? This is called "doubly degenerate".

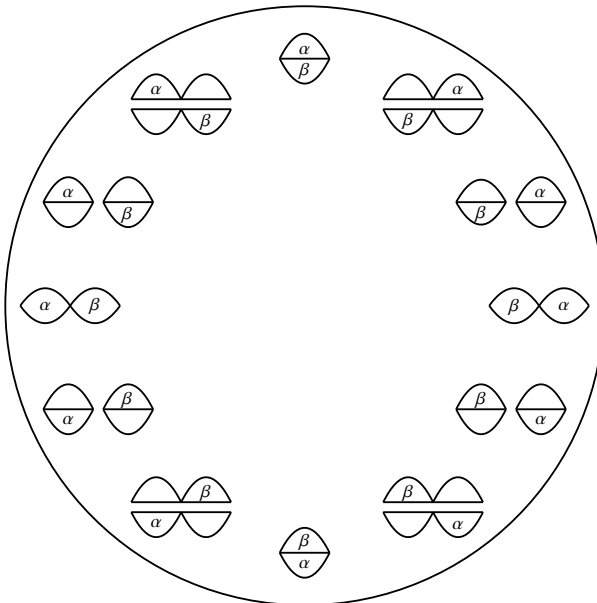
If there is only one 0-cell and only one 1-cell our non-trivial data is now just a set of 2-cells. We now have two types of composition, horizontal and vertical, so we get two binary operations, one coming from vertical composition and the other from horizontal composition. We also know one thing about the interaction between those binary operations, coming from the interchange law.

At this point something rather amazing happens, called the *Eckmann–Hilton argument*. This was originally an argument in algebraic topology used to show that higher homotopy groups are always commutative, but it comes down to a general algebraic principle.

For our doubly degenerate 2-category we consider the interchange law in the following special case, with only two of the 2-cells being non-trivial.



Remember that as the structure is doubly degenerate all the 0-cells are the same, and all 1-cells are identities. We then have the following which I like to call the "Eckmann–Hilton Clock":



The “clock” depicts an argument showing this amazing result: horizontal and vertical composition must be the same binary operation, and moreover, it is commutative. The clock is not a formal proof, but it’s the beginning of one: the idea is that every time you move to the next “hour” on the clock face you are performing a coherence isomorphism. (All the unmarked 2-cells are also identities.)

Things To Think About

T 24.11 See if you can work out what each isomorphism is as you move around this clock. There is some subtlety around 12 o’clock and 6 o’clock, as we have to use a vertical unit horizontally, but everything is strict so it works. (However it becomes critical in weaker situations.) It’s worth also understanding why this argument only works when all the 1-cells are identities.

In summary, we start with a set with two monoid structures on it that are *a priori* different, but satisfy the interchange law. We then show this means that the second monoid structure isn’t really extra structure, but has the effect of forcing the first one to commute. So a doubly degenerate 2-category “is” a commutative monoid.

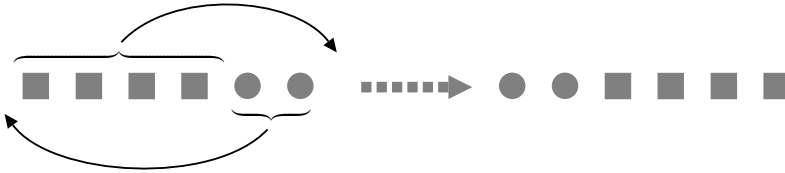
This is a sign of something rather subtle going on in higher dimensions. Higher-dimensional categories are difficult to study for many reasons, and studying various degenerate versions can be a good way to get a handle on subtle coherence questions. As we go up dimensions, the gap between strict and weak versions gets wider and wider. We have seen that for 2-categories the strict and weak versions aren’t *extremely* different. However for 3-categories the gap is wider. Weak 3-categories are called tricategories, and coherence for tricategories is much harder. In particular, it is crucially *not* the case that every tricategory is triequivalent to a strict 3-category. As part of the same phenomenon, it is not the case that every diagram of constraints commutes.

It is much easier to see this effect for *doubly degenerate* tricategories, that is, tricategories with only one 0-cell and only one 1-cell. This is like a categorified version of what we just did for doubly degenerate bicategories. The Eckmann–Hilton argument showed us that doubly degenerate bicategories produce commutative monoids. When we go up a dimension we have a sort of weak Eckmann–Hilton argument showing that doubly degenerate tricategories produce “weakly commutative monoidal categories”.

This is very far from a rigorous definition, but the idea is that if a monoidal category is weakly commutative it is “commutative up to isomorphism”, which means $A \otimes B$ is isomorphic to $B \otimes A$ for any objects A and B . However, as good

category theorists, we prefer to specify an actual isomorphism $A \otimes B \xrightarrow{\gamma} B \otimes A$. Furthermore, that isomorphism then also needs to satisfy some axioms ensuring we can manipulate it as if it were an equality.

This isomorphism γ is called a *braiding*, because it is quite a lot like braiding hair. This idea starts from the idea that commutativity can be shown using physical objects by moving them around each other physically. For example the diagram below depicts $4 + 2$ on the left, and a way of moving the objects past one another to produce $2 + 4$ on the right.



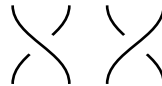
Note that we have to be in at least a 2-dimensional world to do this; if we only have one dimension that means the objects are stuck in a line and won't be able to move sideways. But in a 2-dimensional world we can use the second dimension to get around the first. This is essentially what the Eckmann–Hilton argument was doing in a doubly degenerate bicategory: using the second dimension to get around the first.

Now imagine that those physical objects are sitting on a table but attached to pieces of string hanging from the ceiling. (In the above diagram we are looking at the objects from directly above.) Now when you move the objects around each other the string will get crossed over, or braided. This is how Maypole dancing works.[†]

This is essentially what happens in a tricategory — we can move things past each other but a trace remains of how we did it; we can also think of this as recording the passage of time in a third dimension.

Crucially this also means that if we move the objects past each other the other way, the strings duly cross over the other way, and we get a *different* braid; for example in the above example we could slide the squares round the circles instead in the anti-clockwise direction instead.

We often represent these two possibilities as the two different string crossings shown on the right.



[†] This is a traditional English folk dance where everyone is in a circle and holds onto a long ribbon attached to the top of a tall pole in the center. The dancers weave around each other while going round the pole, making intricate braids in the ribbons. When I was little I was enthralled by how that worked. (I still am.)

Anyone accustomed to braiding hair knows that the under-and-over-crossing methods produce different braids, especially if you're doing a French braid, as I've done in the pictures on the right.



If we were braiding hair in 4-dimensional space we could hop into the fourth dimension to bring one of those strands “through” the other (somewhat like when we do the commutativity in the first place we hop into the second dimension to bring the objects past each other). In 4-dimensional space the braid wouldn't hold, as one strand of the braid could just move through any other strand. In category theory this question arises when we start thinking about degenerate 4-categories.

The *periodic table of n -categories* was proposed by Baez and Dolan as a table showing the sorts of structures that arise with different levels of degeneracy in weak n -categories. It essentially encapsulates increasingly subtle versions of commutativity. At the level of monoids, things are simply commutative or not. With every added dimension there are more senses in which things could be commutative (or not) rather than just “yes” and “no”.

I like to think that studying these lower-dimensional remnants of higher-dimensional structures is like footprints in the snow. I love the book *Miss Smilla's feeling for snow* by Peter Høeg. Miss Smilla is from Greenland and grew up playing a game of deciphering footprints in the snow. One person would close their eyes while another jumped around in the snow making footprints. The first person would then study the footprints and be challenged to reconstruct the actions of the first person. Miss Smilla's understanding of snow is so intimate that she can read the footprints perfectly, even if someone has spun around in the air and landed right back in the same place.

Studying degenerate higher-dimensional categories is like studying the footprints left in the snow by some mysterious higher-dimensional creatures, and helps us to understand how much more subtle they are than 2-categories.

24.9 n and infinity

Higher-dimensional categories are difficult, and get more and more difficult as the dimensions increase. They are already difficult when $n = 3$ and are not very well understood beyond that. In fact, above $n = 3$ they are so difficult that we

typically revert to general n at that point, and don't try to study any particular dimension.

There have been many approaches to defining n -categories, and describing any of them here is far beyond our scope. However, I would like to describe some of the different ideologies behind the approaches.

Here are the general features that a definition needs to have.

Data A way of producing more dimensions. Higher-dimensional morphisms are sometimes called “cells”.

Structure A way of dealing with all the different levels of composition and all the structure/constraint cells that need to exist (and these should all be equivalences at the right level for the codimension).

Properties A way of deciding what axioms the constraint cells should satisfy in order to produce a coherent enough structure. The general principle is that at each dimension there are constraint cells whose interaction is mediated by constraint cells at the dimension above; this continues to the top dimension where there is no dimension above, so we need some axioms in the form of equalities. Of course, if we are doing infinity-categories there never is a top dimension so we never have any axioms invoking equalities.

Notation One of the most fundamental problems that makes this all hard is the problem of expressing higher-dimensional cells in the first place. We can't easily draw them any more as we're stuck with our 2-dimensional pieces of paper. Even once we've succeeded in making a definition, doing any sort of calculations or proofs in higher dimensions remains a difficult problem.

Here are some of the ways in which definitions of higher category can take on different flavors.

Iteration vs all-at-once

Some definitions iterate a process of adding a dimension, whereas others start with data giving all dimensions at once. We saw this difference in our two ways of defining 2-categories: one way was by enrichment, where we took the definition of category and added another dimension by using hom-categories instead of hom-sets. The other way involved starting with a 2-globular set as our data, straight off.

Iteration always has the appealing feature that you do one simple thing and then iterate it to produce something very complex. However, for higher categories this idea comes with a complicated feature: as we go up in dimensions, enrichment needs to happen more and more weakly. If we iterate *strict* enrichment we'll only get *strict* n -categories. We can work out how to do weak

enrichment to produce bicategories at the first step, but iterating that will not produce something as weak as tricategories at the next stage (although it will be slightly weaker than a strict 3-category).

Instead of adding in one dimension at a time by iteration we could throw in all dimensions at once. In that case we have to come up with all new ways of dealing with those dimensions. We can't really rely on the existing low-dimensional theory to extend to the higher dimensions, which is perhaps why that branch of research is progressing somewhat slowly. We'll come back to that in "Algebraic vs non-algebraic".

Enrichment vs internalization

Even if we've decided to do iteration, there are (at least) two different flavors of it. One is by enrichment as we've seen, where we replace the hom-sets by hom-categories. The other is called "internalization" and involves the "categorical" definition of category, starting from graphs.

We briefly saw that we could define a category starting from an underlying graph, which is a diagram in **Set** as shown here.

$$C_1 \begin{matrix} \xrightarrow{s} \\ \xrightarrow{t} \end{matrix} C_0$$

In this form of definition, composition and identities are also defined as morphisms in **Set**, and the unit and associativity axioms are expressed as commutative diagrams in **Set**. As this is categorical, we can pick this definition up and place it inside other categories.[†]

This gives us the definition of a category *internal* to another category, and is slightly different from a category *enriched* in another category. For an internal category, the whole underlying graph is now a diagram inside another category, which means something has happened to the objects as well as the morphisms. This is different from enrichment, where the objects are still just a set.

For example, if we take a category enriched in topological spaces, we have a set of objects, and for every pair of objects a space of morphisms between them. If we take a category internal to topological spaces, we have a space of objects and a space of morphisms.

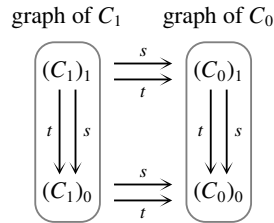
We can also compare categories enriched in **Cat** and categories internal to **Cat**. Categories enriched in **Cat** are 2-categories. When we defined them, we mentioned that the objects did not become a category as they already had morphisms between them. However if we instead take categories internal to categories then the objects are themselves a category before we even think about the morphisms.

[†] The definition of composable pair involves a pullback, so in order to do it we would need to be able to take the relevant pullback in the new category.

The underlying graph as shown on the right is now a diagram in **Cat**, that is, a diagram of categories and functors.

$$C_1 \begin{matrix} \xrightarrow{s} \\ \xrightarrow{t} \end{matrix} C_0$$

This means that C_1 and C_0 themselves have underlying graphs. If we unravel that, we have the diagram in **Set** shown on the right. You might find yourself getting a bit dizzy trying to understand all the parts of this; if so that's understandable.



In this structure the 0-cells $(C_0)_0$ have two types of 1-cell between them:

- $(C_0)_1$: those coming from the original category of objects C_0 , and
- $(C_1)_0$: the objects of the category of morphisms C_1 .

These are usually depicted as horizontal and vertical arrows. We then have 2-cells given by $(C_1)_1$ and these have a source and target in $(C_1)_0$ and also a source and target in $(C_0)_1$, with some commuting conditions.

The end result is that 2-cells have this shape; remember that the horizontal and vertical 1-cells come from different parts of the abstract structure and can't be composed: they're different.

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ p \downarrow & \Downarrow \alpha & \downarrow q \\ c & \xrightarrow{g} & d \end{array}$$

Here f and g are the source and target of α in $(C_1)_0$ and p and q are the source and target of α in $(C_0)_1$. We could have picked the horizontal and vertical conventions the other way round, but I like it this way as it shows the generalization from globular cells vividly: globular 2-cells can be regarded as square ones in which all the vertical 1-cells are identities (collapsing the sides to a point). This comes from the category C_0 being discrete, that is, $(C_0)_1$ being trivial; the underlying data then collapses to a 2-globular set.

This structure with square-shaped 2-cells is called a *double category*. It still has horizontal and vertical composition and interchange. The square-shaped cells complicate the situation but turn out to be quite efficacious for addressing some questions in algebraic topology. When this process is iterated we get n -dimensional cubes at each dimension, and indeed these types of n -categories are called cubical rather than globular.

Things To Think About

T 24.12 It's quite interesting to try fully unraveling all the definitions to see where the structure of a double category comes from. You could start by checking that my diagram of a 2-cell makes sense — how do we know the endpoints of the source and target 1-cells need to match up like that? The key is to think about the fact that the source and target maps in the underlying graph are func-

tors. So on underlying data they are morphisms of graphs, thus some commuting conditions arise in the unraveled diagram.

Algebraic vs non-algebraic

Instead of building dimensions by iteration we can start with all dimensions of data at once, with an underlying infinite-dimensional globular set or something similar. We've seen from the previous discussion that there are different ways of expressing the underlying data: it could be a globular set, or we could have cells of different shapes such as cubes. There are in fact many other different underlying shapes that are possible, and they are usually described by using a different adjective on the word "set". Thus so far we have had globular sets and cubical sets, but there are also simplicial sets (which have been well studied by topologists for some time) and then shapes called things like "opetopic", "multitopic", "dendroidal", "multisimplicial".

The shapes are not just arbitrary choices; they're related to the way in which we then go on to express the structure on the underlying data. In some cases the definition involves specifying things like binary operations at every dimension. This is called *algebraic*, because algebra is about operations combining multiple inputs into one output. In higher-dimensional algebra we often need to combine more than two inputs, so we might generalize to "*k*-ary operations" with *k* inputs.

These generalized operations are often expressed by means of monads. We have not really talked about monads but they are special functors which are particularly good for generating algebraic structures "freely", as we did for the free monoid construction. Once we have generated the structure freely we then say "right, now we want this structure actually to have a value in our underlying data". So for a monoid we would start with an underlying set *A*, generate the free monoid on it (all the words) and then say: given any word *abc*, say, I want this actually to have a value in my original underlying set. For example in the natural numbers we would say " $2 \times 4 \times 3$ is a valid possible multiplication, now what does it actually equal?"

Abstractly this amounts to giving a function $\text{free monoid on } A \longrightarrow A$.

This is something that we can do using *algebras for monads*. (I'm giving the name just so that you can look it up elsewhere if you are interested.) What we do for *n*-categories is something like this: start with an underlying globular set *A*, generate the free *n*-category on it, and then evaluate all those operations by means of a morphism (of globular sets) $\text{free } n\text{-category on } A \longrightarrow A$.

A key feature of expressing structure algebraically is that we have definite

answers to things. We know that $2 \times 4 \times 3$ definitely equals 24 in the natural numbers. However, there are situations in which answers are less definite. For example, suppose that instead of doing multiplication of natural numbers we are doing products of sets. We know that a product of three sets *exists* but there are many possible candidates for it. So instead of having one definite answer we have many possible answers, and they are all uniquely isomorphic.

This idea is expressed by *non-algebraic* approaches to n -categories. To express structure non-algebraically we don't actually specify what the results of operations are, we just verify that there is enough structure around making valid possible results exist. For example, composition might be possible in various ways; even in a weak 2-category we have two different composites of this diagram $\boxed{a \longrightarrow b \longrightarrow c \longrightarrow d}$ because we don't have strict associativity.

Non-algebraic approaches typically use shapes of structure more complicated than globular cells, in order to be able to build in enough scope for expressing what the candidates for composites are. Arguably, the approaches based on simplicial sets have made by far the most progress. Simplicial sets are based on "simplices", which are higher-dimensional generalizations of triangles. In three dimensions the shape in question is a tetrahedron, which we saw in Section 8.6 is a shape that gives the geometry of associativity for three composable arrows. In n dimensions the shape gives the geometry of associativity for n composable arrows.

The simplicial approaches are greatly helped by a huge body of existing work on simplicial sets from the fields of topology and homotopy theory. I think this accounts for why non-algebraic approaches have made more progress than algebraic approaches: homotopy theorists have long been used to asking for structure to exist rather than worrying about exactly what it is, and have developed many techniques for dealing with it. However, algebraic approaches are not made redundant by this; in fact I think it's particularly important to remember the algebraic questions that are left unanswered by the non-algebraic approaches.

Finite dimensions vs isomorphisms forever

We have seen that a set "is" a category with no non-trivial morphisms, that is, all morphisms are identities. If we relax that condition slightly we get a category in which all the morphisms are *isomorphisms*. This is the definition of a groupoid.

We could do this one dimension up, to express categories as 2-categories: we could say that a category "is" a 2-category in which all 2-cells are identities, and then we could relax that and say that all 2-cells are isomorphisms.

But as is typical in higher-dimensional category theory we could now say “Why stop there?” Whenever we cut off at n dimensions, we are essentially saying that every morphism at a dimension above n is the identity. Instead, we could say that every morphism above dimension n is an isomorphism, or rather, an equivalence at the appropriate level. If we decide we’re never going to cut ourselves off, we will keep doing this forever into infinite dimensions. We will get an infinity category in which every morphism above dimension n is an equivalence, for an appropriate infinite-codimensional[†] notion of equivalence.

Things To Think About

T 24.13 You might like to see if you can create that definition of equivalence for yourself. Start by recalling how we went from inverses to pseudo-inverses. Now find the place in the definition of pseudo-inverse that involves inverses, and replace those by pseudo-inverses. And then keep going because there is no top dimension, so there is no point where we should talk about equalities between k -cells.

These are sometimes called (∞, n) -categories. They arise naturally in topology because topological spaces effectively have no top dimension — there is always another level of homotopy that can be defined. But sometimes one wants to focus on a few dimensions at the bottom and sort of let the others take care of themselves, hence we look at n dimensions closely but without having to chop off and “seal up” the top dimensions.

One of the advantages of this approach is that you can do everything maximally weakly right from the start, which might help with iteration. You don’t start with equalities that you then have to weaken to isomorphisms when you go up a dimension (and so on), because you had infinite dimensions all along. However, only a finite number of the dimensions are really non-trivial, so you don’t end up with the full complications of infinite dimensions. This approach has been particularly fruitful in non-algebraic definitions, as a certain amount of non-specified-ness is helpful for allowing all those equivalences to whiz around all the way up to infinity. The theory of $(\infty, 1)$ -categories is particularly well developed, so much so that they are sometimes called “infinity-categories”.

Having infinite dimensions might sound much more complicated than strictly having n , but I proved a theorem[‡] which shows that there are some advantages:

[†] Remember “codimension” is the number of dimensions *above* a cell, so if we’re in an infinity-category then all cells have infinite codimension.

[‡] An ω -category with all duals is an omega groupoid. In *Applied Categorical Structures*, 15(4):439–453, 2007.

you never have to worry about specifying axioms at the top dimension to seal up the ends, because there is no top dimension. I like to think it's the fact that if we were immortal we could procrastinate forever.

24.10 The moral of the story

Higher-dimensional category theory is currently still developing as a field of research. It is difficult because of the infinite levels of nuance and subtlety that can arise, and because of the difficulties inherent to representing infinite dimensions in our very limited three-dimensional physical world.

The point of doing it is something like going up high to get a better overall view of a situation. I'd like to end by coming back to the analogy I made in Chapter 2 about shining a light. I like the analogy for abstraction in general, and it is relevant to higher-dimensional category theory specifically. As I said, if you're shining a light on something to see it better you can raise the light higher up to illuminate a broader area and see more context, but you might see less detail. If you shine the light very close it will be very bright but you will lose context. This is a trade-off we make whenever we get involved with abstraction and higher dimensions. We gain overview and context. We gain the chance to unify a wider variety of structures. We see more connections between different areas. But we also lose detail. In the case of abstraction we lose specific detail about specific situations, and in the case of higher dimensions we often lose our grasp on detail, or our ability to grasp the detail, because the complications are so extreme. So while I am very drawn to the higher-dimensional approaches it's also important to have ways to operate in lower dimensions, by finding ways to temporarily ignore the extra nuance of the higher dimensions without ruining too much. Basic 1-dimensional category theory has proved itself to be a particularly good level for balancing those aims in much of mathematics, and beyond.

For me there is one final, even more abstract attraction of infinite-dimensional category theory: I think of it as a fixed-point of abstraction. If science is the theory of life, and mathematics is the theory of science, and category theory is the theory of mathematics, I think that 2-category theory is the theory of category theory, and 3-category theory is the theory of 2-category theory, and so on. Finally, however, higher-dimensional category theory is the theory of higher-dimensional category theory. We have reached a pinnacle of abstraction. Or perhaps the heart of abstraction, or its deepest roots.

Perhaps the pinnacle and the heart and the deepest roots are the same, and that, to me, is the joy of abstraction.

Epilogue

Thinking categorically

In case you have become overwhelmed with details and technicalities, we will end by taking a step back and summing up some of the main principles of category theory.

Mathematics is founded on rigor and precision. The discipline relies on a large quantity of formality in the form of notation, terminology, and use of language that is slightly different from the use of language in everyday life. This is one of the crucial things that gives math its clarity and thus its power, but learning this formality can get in the way of feeling its spirit. I think this is like learning the piano and worrying so much about playing the right notes that you don't enjoy the music itself. But at the other extreme if you never worry about playing any right notes there are some things you will never be able to play.

Most math textbooks focus on the formality and have less focus on the spirit, with face-to-face teachers often being the ones to convey the spirit that the text couldn't, or didn't. The thing is that math can then end up seeming like a compilation of a series of very technical processes. The more abstract the math is, the more it can seem like that, which is a particular risk for category theory. And it is then a risk that the *practice* of category theory actually does become a series of very technical processes, if that is how it is conveyed by textbooks, and thus that is how it is learned. In this book I have wanted to convey some of the technicality and a great deal of the joy. Because for me the act of doing category theory feels more like soaring than taking small technical steps, at least at the beginning; the small technical steps come later when you're making sure everything is truly rigorous.

So I wanted to end by summing up what I think it means to “think categorically” rather than just “do category theory”. I want to gather together the various principles of category theory that I've mentioned as we're going along, so that we can end with spirit rather than with technicalities.